

House Price Prediction Model

Project 4 - Machine Learning Regression Analysis

This project predicts house prices using Machine Learning techniques. The Housing dataset was analyzed using Exploratory Data Analysis (EDA), One-Hot Encoding, Multi-Collinearity Analysis, Linear Regression, and Ridge Regression.

Dataset Information

Dataset contains 545 records and features including area, bedrooms, bathrooms, stories, parking and several categorical attributes.

Data Preprocessing

Missing values were checked. Categorical features were transformed using One-Hot Encoding as required in the project statement.

Figure 1: Distribution of Numerical Features

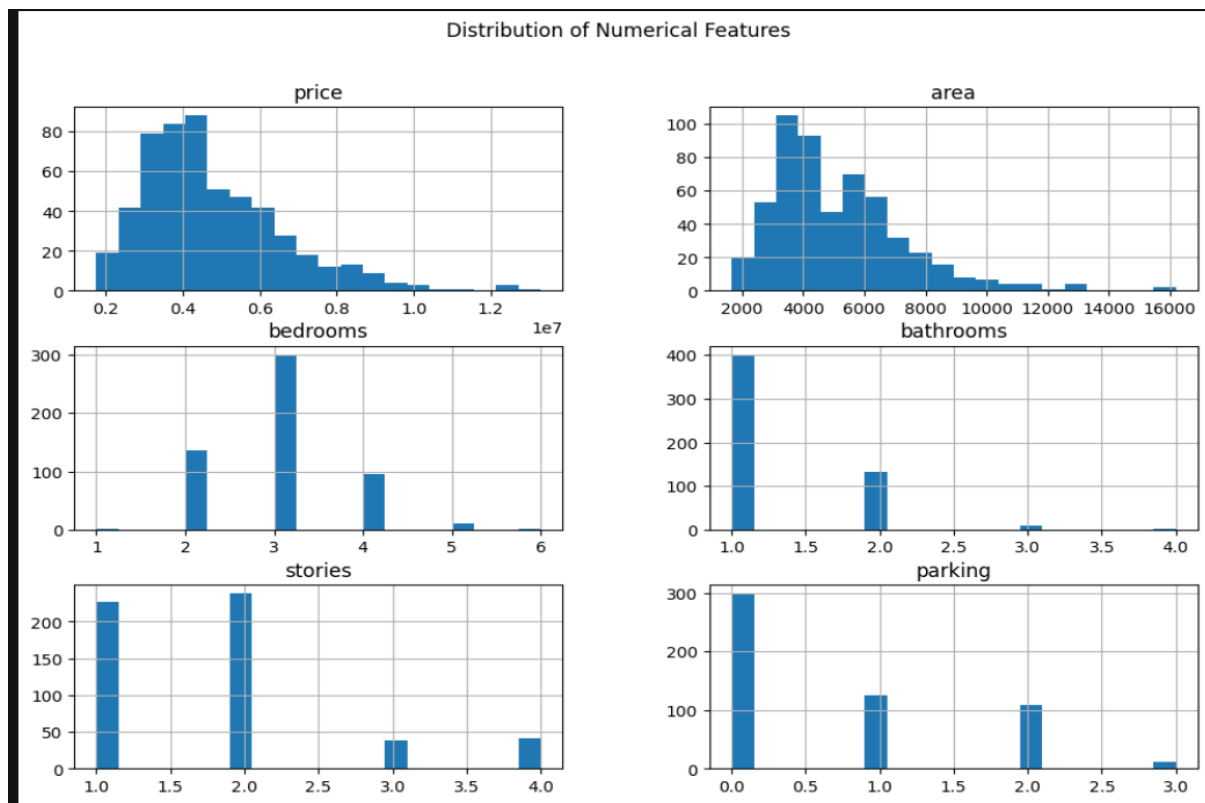


Figure 2: Outlier Detection using Boxplot

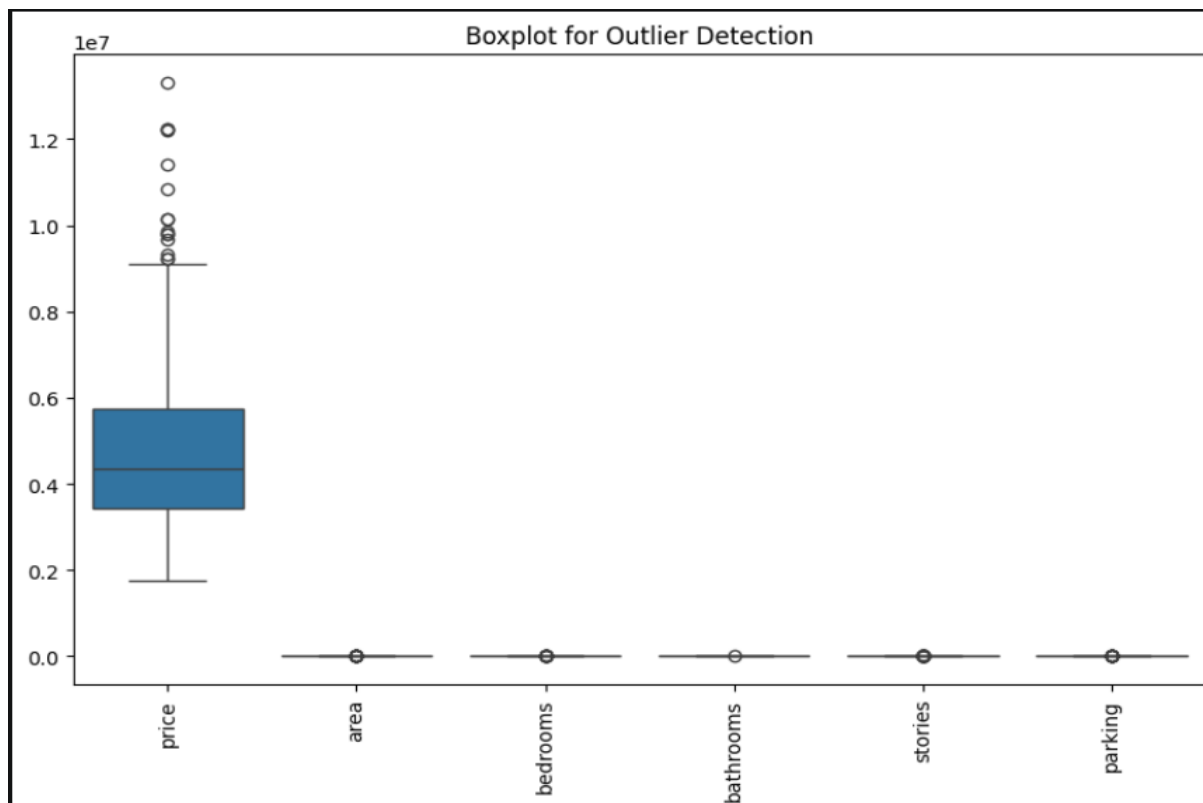


Figure 3: Correlation Heatmap

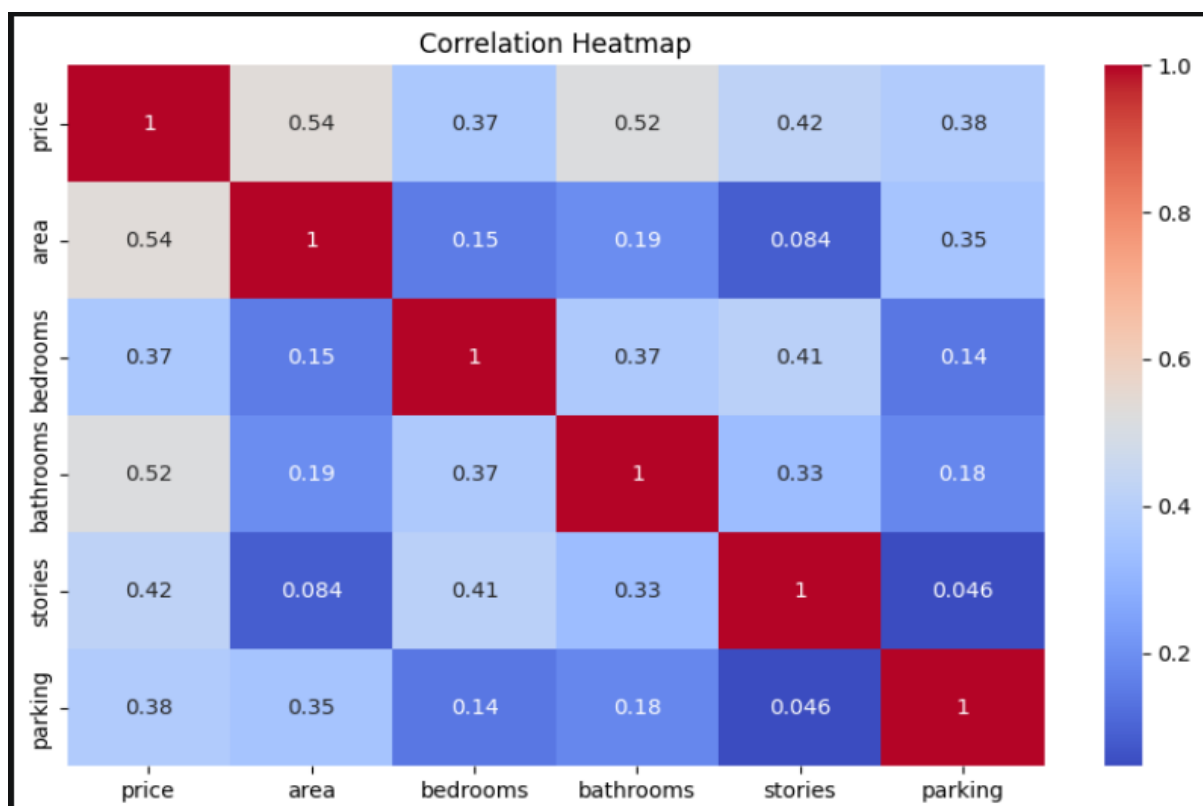


Figure 4: Bedrooms vs Price Analysis

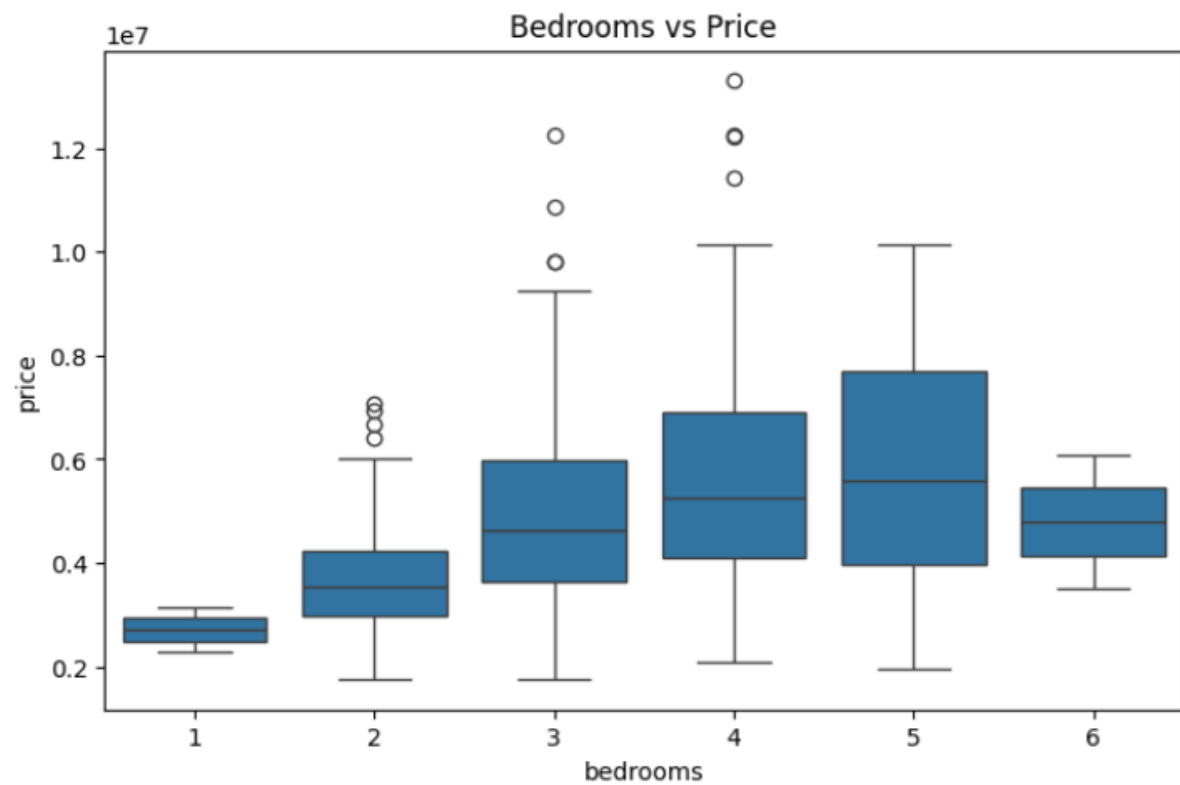


Figure 5: Area vs Price Scatter Plot



Model Building

Linear Regression and Ridge Regression models were trained on the processed dataset. Performance was evaluated using MAE, RMSE and R^2 Score.

Results Summary

Linear Regression R^2 Score ≈ 0.653

Ridge Regression R^2 Score ≈ 0.652

VIF analysis was performed to check multi-collinearity.

Linear Regression performed slightly better than Ridge Regression.

Conclusion

The project successfully satisfies all requirements: Missing value inspection, One-Hot Encoding, Multi-Collinearity Check, Predictive Model Development, and Error Analysis (MAE/RMSE). The final model demonstrates a strong foundation in Regression Analysis.